2nd OpenFOAM HPC Challenge (OHC-2)

JULY 1, 2026

Mini-symposium to be held in the auspices of the
21st OpenFOAM Workshop, Guimarães, Portugal

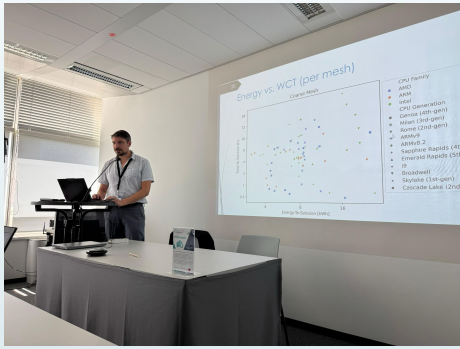
Organized by the OpenFOAM HPC Technical Committee (TC)

OpenFOAM HPC Challenge (OHC)

Background

- The **OpenFOAM HPC challenge** is an annual event that runs in conjunction with the OpenFOAM workshop
- The event brings together **hardware vendors, industrial users and developers** to present recent advances in optimizing OpenFOAM performance
- Participants submit simulation results obtained with various **software enhancements**, and on **state-of-the-art hardware**
- Results are **analyzed, discussed** and **disseminated** to the community:
  https://arxiv.org/abs/2603.27565

  github.com/OpenFOAM-HPC-Challenge





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Objectives

- Evaluate the **efficiency and scalability** of OpenFOAM on different hardware platforms (hardware track)
- Compare the performance of latest OpenFOAM **code enhancements** and flavors (software track)
- Propose **unified metrics** that will help guide the future development of OpenFOAM on next-generation hardware
- Bring together **hardware vendors, industrial users and developers** to discuss OpenFOAM performance
- **Publish benchmark performance** results of OpenFOAM on state-of-the-art hardware for guidance to users and developers

2nd OpenFOAM HPC Challenge (OHC-2)

Scope

- The focus of this mini-symposium will be the industrial testcase of external flow over a **static DrivAer automotive model**, investigated as case 2 in the AutoCFD2 workshop, and as case 2a in the subsequent AutoCFD3 & AutoCFD4 workshops
- A **full OpenFOAM setup*** of the testcase, including meshes, physical model, and solver selection can be found here
- **Reference numerical results** shall also be made available soon
- **In the hardware track**, participants are invited to submit and present results obtained with the **pre-defined setup using an official OpenFOAM version**, on any combination of hardware
- **In the software track**, participants are invited to submit and present results obtained with any OpenFOAM version and custom solver, on any hardware, so long as the mesh and physical modelling remains true to source, and accuracy is retained
- Participants are also invited to submit and present HPC enhancements related to **pre- and post-processing** of OpenFOAM data, applied to the above testcase

*Special thanks to Upstream CFD for providing the OpenFOAM DrivAer setup, which has been adapted for the challenge



2nd OpenFOAM HPC Challenge (OHC-2)

Timeline

- **March 2026: Call for Participation + Release of test case and ref. results**
- **April 2026: Virtual Q&A meeting (participants + organizers)**
- **April 2026: Notice of intent to participate (e-mail)**
- **May 31, 2026: OpenFOAM Workshop regular registration deadline***
- **June 14, 2026: Data submittal deadline**
- **July 1-2, 2026: Mini-symposium (part of OpenFOAM workshop)**
- **August 1, 2026: Release of full performance data**

*** Registration to OFW is required for participating in-person in the OHC-2 mini-symposium**

2nd OpenFOAM HPC Challenge (OHC-2)

Rules (Hardware Track)

- **Objective:** Demonstrate and compare the performance of OpenFOAM on various hardware configurations

- **Hardware:** unconstrained*, so long as it supports standard OpenFOAM
- **Software**: OpenFOAM.com v2512
- **Compiler**: unconstrained
- **MPI**: unconstrained
- **Case setup**: occDrivAerStaticMesh, with fixed mesh, solver, decomposition method and numerical models (no changes allowed except for no. of processors for decomposition and coefficients for decomposition method)
- **Mandatory** pre-processing stages (decomposition, renumbering, potentialFoam init)

*** Access to hardware may be applied through:**
EuroHPC JU Benchmark Access call

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Rules (Software/Open Track)

- **Objective**: Showcase code/hardware optimizations that improve performance (time, energy), whilst retaining the basic level of accuracy as demonstrated with standard OpenFOAM
- **Hardware**: unconstrained*
- **Software**: unconstrained, OpenFOAM-based code
- **Case setup**: occDrivAerStaticMesh, with fixed mesh and physical modelling (temporal and spatial discretization, turbulence model)
  - Steady-state, incompressible flow
  - RANS with kw-SST turbulence model
  - Pre-defined fvSchemes (changes allowed but have to be reported)
- **Proposed topics for investigation**: accelerators, pre-/post-processing (I/O), mixed-precision, linear solvers, renumbering/decomposition

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EuroHPC JU Benchmark Access call

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Metrics

- **Hardware/Software Track**
  - Single- and Multi-node Scalability
  - Time-to-solution, excl. Pre-processing (decomposition, renumbering) and initialization (potentialFoam)
  - Energy-to-solution
  - Accuracy w.r.t reference solution obtained with standard OpenFOAM
    - Aerodynamic coefficients

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Administration

- Questions/issues can be reported in the Forum or via e-mail
- Submissions via Dropbox
- Mini-symposium schedule
  - **July 1-2**
- Speaker slots (15 minutes) will be allocated to all participants who submit at least one contribution (on-site/remote)
- Limited number of slots available for online presentation, for those who cannot join in-person (please notify in advance)

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Organizing Committee

- Mark Wasserman, **NVIDIA**
- Sergey Lesnik, **Wikki GmbH**
- Gregor Olenik, **TUM**
- Ivan Spisso, **Leonardo Company**
- Charles Mockett, **Upstream CFD GmbH**
- Fillipo Spiga, **NVIDIA**
- Neil Ashton, **NVIDIA**
- Elisabetta Boella, **E4 Computer Engineering**
- Fabrizio Magugliani, **E4 Computer Engineering**
- Simone Bna, **CINECA**
- Gavin Tabor, **Univ. of Exeter**

2nd OpenFOAM HPC Challenge (OHC-2)

Data Submission Guidelines (General)

- To maximize the outcome of the workshop for the OpenFOAM community, a structured simulation post-processing approach was adopted. The structured approach will ensure that:
  - Simulation results can be compared to reference results
  - Solutions from different participants can be compared
  - The organizing committee will be able to process all incoming data
- To enable a structured data processing all participants are requested to use the provided spreadsheet to submit their results:
  - Detailed hardware/software setup
  - Wall-clock measurements
  - Aerodynamic forces (mean + history)
- In case results are not provided in the required format, the submission will not be considered for comparison, and a speaker slot can not be guaranteed

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Data Submission Guidelines (General)

- Participants are required to submit one spreadsheet per mesh and track
  - Please don't mix data from multiple mesh resolutions or HW/SW track in one submission
- Multiple submissions are allowed
- Structure of the Excel template*
  - **META Data**: Description of simulation environment (SW/HW)
  - **Simulations:** Timings of multiple simulations for the same mesh/track
  - **Aero Forces**: Mean Aerodynamic Forces + time/iteration history
  - **Force Plots**: Generated plots of aerodynamic forces + reference results
  - **Convergence Plots:** Generated plots of convergence histories for selected simulations

How to extract data required for submission?

Example (full) submission

*Special thanks to Burkhard Hupertz for providing the original (AutoCFD4) data submission template, which has been adapted for OHC-1

2nd OpenFOAM HPC Challenge (OHC-2)

Data Submission Guidelines (Software Track)

- What details need to be reported regarding custom solvers in the software track?
  - Numerical schemes
  - Linear solver
  - Statistical analysis of aerodynamic forces
  - (optional) input file
  - (optional) Meancalc analysis
  - Precision
- All data submitted will be made available to participants after the workshop via a dedicated repository
- Submitted data will be used for comparison, and compilation of a manuscript to be submitted to the OpenFOAM journal